

GreenHarvest Agro Pvt. Ltd.

Human Capital Management (HCM) | Production Planning (PP) | Warehouse Management (WM)

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Chapter 1 | Abstract

This project report presents a comprehensive implementation blueprint for deploying SAP ERP (Enterprise Resource Planning) at GreenHarvest Agro Pvt. Ltd., a fictitious but realistically modelled agricultural and food-processing enterprise headquartered in Bangalore, India. The blueprint focuses on three interdependent SAP modules: Human Capital Management (HCM), Production Planning (PP), and Warehouse Management (WM). Together, these modules address the organisation's critical operational challenges — workforce administration, production scheduling, and inventory control — within a unified digital ecosystem.

The primary objectives of this project are: (i) to design the enterprise organisational structure within the SAP environment; (ii) to detail module-specific configuration steps using SAP's Project Reference Object (SPRO) customisation tool; (iii) to map cross-module integration points; and (iv) to propose a phased implementation strategy aligned with industry best practices. The report also addresses risk mitigation, change management, and future-state technology enhancements such as AI-driven analytics and SAP S/4HANA cloud migration.

SAP ERP remains the global benchmark for large and mid-size enterprises seeking operational excellence, compliance, and real-time data visibility. By deploying HCM, PP, and WM, GreenHarvest Agro Pvt. Ltd. is positioned to eliminate manual inefficiencies, reduce production downtimes, and achieve end-to-end traceability across its supply chain — from farm input procurement to consumer distribution.

Chapter 2 | Introduction to SAP ERP

2.1 What is SAP ERP?

SAP (Systems, Applications & Products in Data Processing) is a German multinational software corporation that develops enterprise software solutions used by over 440,000 customers across 190 countries. SAP ERP is an integrated suite of business applications that enables organisations to manage core business processes — finance, human resources, production, logistics, and procurement — through a centralised, real-time database. First introduced as R/2 in 1979 and subsequently evolved into R/3 and then SAP ERP ECC 6.0, the platform is now transitioning organisations towards SAP S/4HANA, its next-generation in-memory ERP solution.

At its core, SAP ERP operates on the principle of a single source of truth — all business transactions are recorded in one integrated system, thereby eliminating data silos, reducing redundancy, and enabling cross-functional reporting. The modularity of SAP ERP allows organisations to implement only the modules relevant to their operations, making it scalable for businesses of all sizes.

2.2 Importance in Modern Businesses

In today's highly competitive business landscape, organisations require real-time insights, process automation, and regulatory compliance across all operational areas. SAP ERP delivers these capabilities through:

- Real-time data processing and reporting, enabling immediate decision-making.
- Process standardisation across departments, reducing errors and improving consistency.
- Regulatory compliance tools embedded within each module (e.g., statutory payroll, GST in India).
- Seamless integration between functional departments, eliminating manual data transfer.
- Scalability to support business growth without replacing the core system.

2.3 Overview of Selected Modules

Module	Full Name	Primary Function	Key Transactions
HCM	Human Capital Management	Manage employee lifecycle, payroll, and org structure	PA40, PA30, PY00, PO10
PP	Production Planning	Plan, schedule, and control manufacturing operations	MB01, CO01, CR01, CS01
WM	Warehouse Management	Manage warehouse operations, storage bins, and inventory	TR01, L50A, LS01, LI01

These three modules are not standalone; they interact continuously during daily business operations. HCM provides the workforce assigned to production work centres (PP), while PP triggers material movements that are physically managed by WM. This tight integration is one of the key strengths of SAP ERP as deployed at GreenHarvest Agro Pvt. Ltd.

Chapter 3 | Company Overview – GreenHarvest Agro Pvt. Ltd.

3.1 Corporate Profile

Company Name	GreenHarvest Agro Pvt. Ltd.
Industry Sector	Agriculture & Food Processing
Registered Office	Electronics City, Phase II, Bangalore – 560100, Karnataka, India
Year of Incorporation	2010
Annual Turnover	INR 420 Crores (FY 2023–24)
Employees	Approx. 1,850 (including seasonal workforce)
Certifications	ISO 22000:2018, FSSAI Licensed, Organic India Certified
Key Products	Organic Cereals, Cold-Pressed Oils, Packaged Pulses, Ready-to-Cook Kits

3.2 Business Model

GreenHarvest Agro Pvt. Ltd. operates a vertically integrated business model spanning farm-level procurement, in-house processing, quality assurance, and pan-India distribution. The company sources raw agricultural produce (wheat, rice, pulses, oilseeds) directly from a network of over 3,200 registered farmer-partners across Karnataka, Andhra Pradesh, and Maharashtra under its 'GreenLink Farming Initiative'. This model ensures traceability, consistent quality, and fair pricing to farmers while maintaining competitive production costs.

The food-processing division operates two manufacturing plants — one in Bangalore (primary) and one in Pune (secondary) — equipped with sorting, milling, cold-press extraction, and packaging lines. Finished goods are distributed through three distribution channels: modern trade (retail chains), e-commerce (own website and marketplaces), and institutional buyers (hotels, hospitals, government procurement).

3.3 Operational Workflow (Pre-SAP)

Prior to SAP implementation, GreenHarvest relied on a combination of standalone software (Tally for accounts, Excel for production scheduling, paper-based warehouse records), resulting in significant operational challenges:

- Production delays due to lack of real-time material availability data.
- Payroll discrepancies caused by manual attendance and overtime calculations.
- Inventory discrepancies between physical stock and recorded stock (average variance: 12%).
- Inability to track batch-level quality attributes from raw material to finished goods.
- Delayed MIS reports, hampering executive decision-making.

3.4 Rationale for SAP ERP Adoption

Given the scale of operations, regulatory requirements (FSSAI, GST), and competitive pressure, GreenHarvest's management decided to implement SAP ERP ECC 6.0 with a phased roadmap towards S/4HANA migration. The three modules — HCM, PP, and WM — were identified as the highest-priority implementation areas based on a business impact assessment conducted by an internal process excellence team.

Chapter 4 | Organisational Structure Design

The SAP organisational structure forms the foundational skeleton upon which all business transactions are executed. Each module has its own hierarchy of organisational units, and these units are interlinked to reflect GreenHarvest's actual business architecture.

4.1 HCM Organisational Structure

The HCM organisational structure in SAP defines how the company is divided for HR administration, payroll, and time management purposes. It comprises the following key elements:

Organisational Unit	SAP Object	Description for GreenHarvest
Client	Client 100	Highest-level SAP unit; represents the entire corporate group.
Company Code	GH01	Legal entity – GreenHarvest Agro Pvt. Ltd.; used for all FI postings.
Personnel Area	GHBA (Bangalore)	Represents the Bangalore plant HR area; separate area for Pune (GHPU).
Personnel Subarea	PROD, WRHS, ADMN	Differentiates production shop floor, warehouse, and admin staff.
Employee Group	Regular, Seasonal	Distinguishes permanent and seasonal workforce.
Employee Subgroup	Hourly, Salaried	Determines payroll schema and time evaluation rules.
Org Unit	OU_HR, OU_OPS	Functional departments: HR, Operations, Finance, QA, Logistics.
Position	Various	Specific job seats: Plant Manager, Warehouse Supervisor, etc.
Job	Various	Generic job titles used across positions (Manager, Operator, etc.)

Figure 4.1: HCM Organisational Hierarchy – GreenHarvest Agro Pvt. Ltd.

Client 100 → Company Code GH01 → Personnel Area GHBA / GHPU → Personnel Subareas (PROD | WRHS | ADMN) → Employee Groups (Regular | Seasonal) → Employee Subgroups (Hourly | Salaried) → Org Units → Positions → Employees

Figure 4.1: HCM Organisational Structure of GreenHarvest Agro Pvt. Ltd.

4.2 PP Organisational Structure

The Production Planning organisational hierarchy defines where and how production takes place within the company. Key organisational elements for GreenHarvest are:

Organisational Unit	SAP Object	Description for GreenHarvest
Client	Client 100	Shared with HCM and WM.
Company Code	GH01	Same legal entity as HCM.
Plant	GH10 (Bangalore), GH20 (Pune)	Each plant has independent production resources.

Organisational Unit	SAP Object	Description for GreenHarvest
Storage Location	SL01 – Raw Material, SL02 – WIP, SL03 – Finished Goods	Physical locations within the plant for inventory management.
Work Centre	MILL01, PRESS01, PACK01	Machines/production lines: Milling, Cold-Press, Packaging.
MRP Area	GH10_MRP	MRP planning segment for Bangalore plant.
Production Version	PV01, PV02	Alternate BOM/routing combinations for different batch sizes.

Figure 4.2: PP Organisational Structure – GreenHarvest Agro Pvt. Ltd.

Client 100 → Company Code GH01 → Plants (GH10 | GH20) → Storage Locations (SL01 Raw | SL02 WIP | SL03 FG) → Work Centres (MILL01 | PRESS01 | PACK01) → MRP Areas → Production Versions

Figure 4.2: PP Organisational Structure of GreenHarvest Agro Pvt. Ltd.

4.3 WM Organisational Structure

Organisational Unit	SAP Object	Description for GreenHarvest
Warehouse Number	WH10	Central warehouse at Bangalore plant.
Storage Type	001 – Bulk, 002 – Rack, 003 – Cold Storage, 004 – Staging	Different physical storage zones within WH10.
Storage Section	A – Fast-Moving, B – Slow-Moving	Subdivisions of each storage type for optimised putaway.
Storage Bin	001-A-001 to 001-Z-099	Granular bin addresses within each storage section.
Quant	System-generated	Represents the actual stock quantity sitting in a specific bin.
Movement Type	101, 201, 261, 311	Goods receipt, issue, production consumption, transfer.

Figure 4.3: WM Organisational Structure – GreenHarvest Agro Pvt. Ltd.

Client 100 → Plant GH10 → Warehouse WH10 → Storage Types (001 Bulk | 002 Rack | 003 Cold | 004 Staging) → Storage Sections (A Fast-Moving | B Slow-Moving) → Storage Bins → Quants

Figure 4.3: WM Organisational Structure of GreenHarvest Agro Pvt. Ltd.

Chapter 5 | SAP Module Explanation

5.1 Human Capital Management (HCM)

5.1.1 Employee Lifecycle Management

SAP HCM manages the entire employee lifecycle from recruitment through separation. At GreenHarvest, the lifecycle encompasses: candidate application (integration with SAP E-Recruiting), hiring action (PA40 – Personnel Action), master data maintenance (PA30 – Maintain HR Master Data), training assignments (Training & Event Management), performance appraisals, and off-boarding with full settlement calculation. Each lifecycle event triggers workflows that update downstream records in payroll, time management, and organisational management.

5.1.2 Payroll Processing

GreenHarvest's payroll is processed using SAP Payroll (India localisation schema INI). The payroll run encompasses basic pay, house rent allowance (HRA), special allowances, overtime wages (for hourly workers), PF (Provident Fund), ESI (Employee State Insurance), professional tax (PT), and income tax under TDS. The payroll simulation (PC00_M40_CALC_SIMU) is executed before the live run to validate results. Payslips are generated as Adobe PDF forms and distributed via the Employee Self-Service (ESS) portal.

5.1.3 Organisational Management

SAP Organisational Management (OM) maintains the hierarchical relationships between organisational units, jobs, positions, and employees using infotype 1000–1001. At GreenHarvest, the OM structure is maintained using the Org & Staffing interface (transaction PPOME). Reporting lines, cost centre assignments, and position budgeting are all managed through OM, ensuring that any structural change (e.g., creating a new Warehouse Supervisor position) automatically propagates to downstream HR and Finance processes.

5.2 Production Planning (PP)

5.2.1 Production Planning & MRP

Material Requirements Planning (MRP) is the backbone of SAP PP. At GreenHarvest, MRP is executed using transaction MD01 (Total Planning) on a weekly basis. The MRP run reads the demand from Sales Orders, planned independent requirements, and forecasts, and calculates net requirements after deducting existing stock and open purchase orders. The output is a set of planned orders for in-house production and purchase requisitions for externally sourced raw materials.

5.2.2 Work Centres

Work centres (transaction CR01) represent physical production resources. GreenHarvest's key work centres include MILL01 (Rice/Wheat Milling Unit), PRESS01 (Cold-Press Oil Extraction), SORT01 (Pulse Sorting & Grading), and PACK01 (Automated Packaging Line). Each work centre has defined capacity, scheduling parameters, costing rates, and quality inspection integration.

5.2.3 Bill of Materials (BOM) & Routing

Material	BOM Component	Qty / Base Unit	Component Type
Cold-Press Mustard Oil (1L)	Raw Mustard Seeds	1.35 KG	Raw Material
Cold-Press Mustard Oil (1L)	PET Bottle (1L)	1 EA	Packaging
Cold-Press Mustard Oil (1L)	Label (Mustard Oil)	1 EA	Packaging
Organic Basmati Rice (5KG)	Paddy (Grade A)	6.25 KG	Raw Material
Organic Basmati Rice (5KG)	BOPP Bag (5KG)	1 EA	Packaging
Organic Basmati Rice (5KG)	Silica Gel Sachet	2 EA	Packaging

Routings (transaction CA01) define the sequence of operations and work centres required to manufacture each product. Each operation has assigned standard times (machine time, labour time) which feed directly into capacity planning and production costing.

5.3 Warehouse Management (WM)

5.3.1 Inventory Management

SAP Inventory Management (part of MM module but tightly integrated with WM) records all goods movements at the storage location level. When a goods receipt (GR) is posted for a purchase order (MIGO, movement type 101), the stock quantity increases at the storage location SL01 (Raw Materials). WM then automatically generates a Transfer Order (TO) to physically move the material from the GR zone to the designated storage bin.

5.3.2 Storage Types and Bin Management

GreenHarvest's warehouse WH10 is divided into four storage types: 001 (Bulk Storage – for grain sacks), 002 (Rack Storage – for packaged goods), 003 (Cold Storage – for temperature-sensitive items like oils), and 004 (Staging Area – for goods pending dispatch). Storage sections A and B further subdivide each type for fast-moving and slow-moving items respectively. The putaway strategy is configured as 'Next Empty Bin' for storage type 001 and 'Fixed Bin' for type 002.

5.3.3 Transfer Orders and Physical Inventory

All physical material movements within the warehouse are governed by Transfer Orders (LT01 – Create TO Manually, or LT0A – Create TO from Transfer Requirement). Annual physical inventory is conducted using transaction MI01 (Create Inventory Document), MI04 (Enter Count), and MI07 (Post Differences). Cycle counting is configured for Category A items (high-value materials) on a monthly basis.

Chapter 6 | SAP Customisation Steps (SPRO)

All SAP customisation is performed through the Implementation Guide (IMG) accessed via transaction SPRO. The steps below are detailed navigation paths for configuring the three modules at GreenHarvest Agro Pvt. Ltd.

6.1 HCM Configuration

6.1.1 Enterprise Structure – Company Code

SPRO → Enterprise Structure → Definition → Financial Accounting → Define Company Code

Action: Create Company Code GH01 | Name: GreenHarvest Agro Pvt. Ltd. | Currency: INR

SPRO → Enterprise Structure → Assignment → Human Resources Management → Assign Personnel Area to Company Code

Action: Assign GHBA and GHPU to GH01

6.1.2 Personnel Areas and Subareas

SPRO → Personnel Management → Personnel Administration → Organisational Data → Define Personnel Areas

Action: Create GHBA (Bangalore), GHPU (Pune)

SPRO → Personnel Management → Personnel Administration → Organisational Data → Define Personnel Subareas

Action: PROD (Production), WRHS (Warehouse), ADMN (Administration)

6.1.3 Employee Groups & Subgroups

SPRO → Personnel Management → Personnel Administration → Employee Attributes → Define Employee Groups

Action: 1 = Regular Employee, 2 = Seasonal Worker

SPRO → Personnel Management → Personnel Administration → Employee Attributes → Define Employee Subgroups

Action: H1 = Hourly Regular, S1 = Salaried Regular, H2 = Hourly Seasonal

6.1.4 Payroll Configuration (India)

SPRO → Payroll → Payroll India → Basic Settings → Define Payroll Area

Action: Create Payroll Area GH with weekly/monthly frequency

SPRO → Payroll → Payroll India → Basic Settings → Generate Payroll Periods

Action: Generate monthly periods for FY 2024-25

SPRO → Payroll → Payroll India → Tax → Assign Tax Company to Payroll Area

Action: Link GH01 tax entity to payroll area GH

SPRO → Payroll → Payroll India → Social Insurance → Maintain ESI Settings

Action: Define ESI contribution percentages (Employer: 3.25%, Employee: 0.75%)

6.2 PP Configuration

6.2.1 Plant and Storage Location Setup

SPRO → Enterprise Structure → Definition → Logistics – General → Define, Copy, Delete, Check Plant

Action: Create Plant GH10 (Bangalore), GH20 (Pune)

SPRO → Enterprise Structure → Definition → Materials Management → Maintain Storage Locations

Action: Create SL01 (Raw Material), SL02 (WIP), SL03 (Finished Goods) under GH10

6.2.2 MRP Configuration

SPRO → Materials Management → Consumption-Based Planning → Planning → Define MRP Types

Action: Verify MRP Type PD (MRP with Planned Orders) for production materials

SPRO → Materials Management → Consumption-Based Planning → Plant Parameters → Carry Out Overall Maintenance of Plant Parameters

Action: Set MRP horizon = 90 days, Planning interval = Weekly for GH10

SPRO → Production → Material Requirements Planning → Define MRP Areas

Action: Create MRP Area GH10_MRP linked to Plant GH10

6.2.3 Work Centre Configuration

SPRO → Production → Basic Data → Work Centres → Work Centre Categories → Define Work Centre Categories

Action: Create categories: Machine, Labour, Production Line

SPRO → Production → Basic Data → Work Centres → Define Formulas for Work Centres

Action: Set SAP_001 (Machine Time), SAP_002 (Labour Time) as scheduling formulas

SPRO → Production → Basic Data → Work Centres → Define Capacity Categories

Action: Configure capacity category 002 (Machine) with 20-hour daily availability for GH10

6.2.4 BOM Configuration

SPRO → Production → Basic Data → Bill of Material → General Data → Define BOM Status

Action: Status 01 = Active, Status 02 = Engineering Change

SPRO → Production → Basic Data → Bill of Material → Item Data → Define BOM Usage

Action: Usage 1 = Production, Usage 5 = Sales

6.3 WM Configuration

6.3.1 Warehouse Structure

SPRO → Logistics Execution → Warehouse Management → Master Data → Define Warehouses

Action: Create Warehouse WH10, assign to Plant GH10, Storage Location SL01

SPRO → Logistics Execution → Warehouse Management → Master Data → Define Storage Types

Action: 001 (Bulk), 002 (Rack), 003 (Cold Storage), 004 (Staging Area)

SPRO → Logistics Execution → Warehouse Management → Master Data → Define Storage Sections

Action: A (Fast-Moving), B (Slow-Moving) under each storage type
SPRO → Logistics Execution → Warehouse Management → Master Data → Define Storage Bins
Action: Bulk auto-generate bins 001-A-001 to 001-Z-099

6.3.2 Putaway and Picking Strategies

SPRO → Logistics Execution → Warehouse Management → Strategies → Putaway Strategies → Activate Storage Type Search
Action: For WH10 - activate storage type determination sequence: 004 → 001 → 002 → 003
SPRO → Logistics Execution → Warehouse Management → Strategies → Putaway Strategies → Define Strategy for Storage Type
Action: 001 = Next Empty Bin; 002 = Fixed Bin; 003 = Fixed Bin (temperature-controlled)
SPRO → Logistics Execution → Warehouse Management → Strategies → Picking → Define Strategies for Picking
Action: FIFO (First In First Out) for all storage types

6.3.3 Movement Type Configuration

SPRO → Logistics Execution → Warehouse Management → Activities → Transfers → Define Movement Types
Action: Verify MM movement type 101 (GR from PO) → WM movement type 101
Action: Verify MM movement type 261 (GI to Production Order) → WM movement type 261
Action: Define TO creation: automatic for goods receipts and production issues

Chapter 7 | Integration Between Modules

One of the most significant advantages of SAP ERP is its native integration across all functional modules. For GreenHarvest, the integration between HCM, PP, and WM creates a seamless, end-to-end digital thread from workforce assignment through production execution to warehouse fulfillment.

7.1 HCM ↔ PP Integration

The HCM-PP integration ensures that workforce availability directly informs production capacity planning. Key integration points are:

HCM Event / Data	Impact on PP	SAP Mechanism
Employee attendance data (IT2002)	Updates available labour hours at work centres	Time evaluation → Capacity planning in PP
Shift roster (IT2003)	Defines working time of work centre operators	Work Schedule Rule → Work Centre operating times
Employee leave (IT2001)	Reduces available capacity; triggers capacity leveling	Capacity leveling requirements planning (CM01)
Training completion (LSO)	Qualifies employees for specific production tasks	Qualification link to task list (PP-PI)

7.2 PP ↔ WM Integration

Production Planning and Warehouse Management are tightly coupled at GreenHarvest. Every production order triggers downstream warehouse activities:

PP Event	WM Response	SAP Transaction
Production order release (CO02)	Transfer requirement created for components	HP10 – Create TO from Production Order
Goods issue to production (MIGO – WM)	WM confirms goods picked from raw materials	LT1A – Confirm Transfer Order
Goods receipt from production (MIGO – WM)	WM posts finished goods into storage	LT0A – Auto TO for GR
Production order confirmation (CO15)	Updates production costs; stock valuation	CO15 – Post Goods Movement

7.3 WM ↔ HCM Integration

While the WM-HCM linkage is less direct, it is operationally significant at GreenHarvest, particularly for workforce deployment in the warehouse and for payroll accuracy:

- Warehouse staff attendance (IT2002) determines shift availability for pick/put-away operations.
- Incentive wages for warehouse pickers are calculated based on Transfer Order confirmation quantities recorded in WM, fed back to the payroll via the HR-MM integration framework.
- Headcount data from HCM OM feeds into warehouse capacity reports for annual capacity planning.
- Safety training completion (HCM Training module) is a prerequisite for assigning employees to Cold Storage operations (Storage Type 003) in WM.

Chapter 8 | Business Process Flow

The end-to-end business process at GreenHarvest follows a structured cycle from demand planning through production to finished-goods dispatch. The following describes the key process flows enabled by the SAP HCM-PP-WM implementation.

8.1 Procure-to-Store Process (Raw Materials)

Step	Activity	Detail
Step 1	Demand identification	PP MRP run (MD01) generates purchase requisitions for raw materials based on planned p
Step 2	Purchase Order creation	MM buyer converts PR to PO (ME21N) and assigns vendor from approved supplier list.
Step 3	Goods Receipt	Raw material delivery received at plant; MIGO (Mvt Type 101) posts GR against PO; stock
Step 4	WM Transfer Order	System auto-generates Transfer Requirement; warehouse worker creates TO (LT0A) to pu
Step 5	Quality Inspection	Quality Management (QM) inspection lot created automatically; material placed in QI status
Step 6	Stock Release	QM usage decision (QA11) changes stock to unrestricted; material available for production

8.2 Plan-to-Produce Process

Step	Activity	Detail
Step 1	Demand Planning	Sales order / forecast creates demand signal; PP planner reviews MD04 (Stock/Requireme
Step 2	Planned Order	MRP creates planned order; PP planner converts to Production Order (CO01) and assigns
Step 3	Capacity Check	PP planner runs capacity levelling (CM01); assigns qualified operators via HCM integration
Step 4	Material Staging	WM Transfer Order (LB10) picks raw materials from storage bins and delivers to production
Step 5	Production Execution	Operators execute operations at MILL01 / PRESS01; partial confirmations via CO11N (Ope
Step 6	Goods Receipt – FG	Finished goods received into SL03 (MIGO – Mvt 101 for Production Order); WM auto-puts i
Step 7	Order Settlement	CO88 settles production order; variances posted to cost centre; order status set to TECO (

Figure 8.1: End-to-End Business Process Flow – GreenHarvest Agro Pvt. Ltd.

Demand Planning (PP) → Purchase Requisition (MM) → Goods Receipt (MM+WM) → Quality Inspection (QM) →
Material Staging (WM) → Production Execution (PP) → Finished Goods Receipt (WM) → Dispatch (SD+WM) → Payroll
Close (HCM)

Figure 8.1: Integrated Business Process Flow across HCM, PP, and WM

8.3 Hire-to-Retire Process (HCM)

Step	Activity	Detail
Step 1	Vacancy Requisition	Line manager creates Position requisition via PPOME; HR approves via workflow.
Step 2	Recruitment	E-Recruiting posts vacancy; candidates apply; interviews scheduled via HCM.
Step 3	Hiring Action	PA40 (Personnel Action – Hiring); infotypes IT0000, IT0001, IT0002, IT0007, IT0008 create
Step 4	Time Management	Shift assignment and attendance recording via IT2003, IT2001, IT2002.
Step 5	Payroll Processing	Monthly payroll run (PC00_M40_CALC); payslip distribution via ESS.
Step 6	Performance Appraisal	Annual appraisal via HAP_START_BSP_CLIENT; merit increment uploaded to IT0008.
Step 7	Separation	Leaving action (PA40); settlement calculated; IT0047 (Provident Fund) closed.

Chapter 9 | Implementation Strategy

GreenHarvest Agro Pvt. Ltd. will adopt the SAP ACTIVATE methodology, the standard Agile-based implementation framework recommended by SAP for ECC 6.0 and S/4HANA deployments. The methodology consists of four phases: Discover, Prepare, Explore (Fit-Gap), Realize, Deploy, and Run.

9.1 Phased Deployment Approach

Phase	Duration	Key Activities	Deliverables
Phase 1: Project Initiation	4 weeks	Project charter, team formation, system landscape project plan, RACI Matrix, Kickoff Presentation	Project Plan, RACI Matrix, Kickoff Presentation
Phase 2: Blueprinting (Explore)	8 weeks	Business process workshops, Fit-Gap analysis, SWRS Blueprinting Document, Process Maps	SWRS Blueprinting Document, Process Maps
Phase 3: Realisation	16 weeks	System configuration, SPRO customisation, RFC/BAPI development, Integration Test Results	Configured S/4HANA system, Integration Test Results
Phase 4: Integration & UAT	8 weeks	End-to-end integration testing, User Acceptance Testing, Data Migration Report	UAT Sign-off, Data Migration Report
Phase 5: Go-Live & Hypercare	4 weeks + 8 weeks	Cutover Go-Live, parallel payroll run, hypercare support	Go-Live Checklist, Hypercare Incident Log
Phase 6: Post Go-Live (Run)	Ongoing	System optimisation, user training, change management, Monthly performance reviews, Enhancement log	Monthly performance reviews, Enhancement log

9.2 Risk Management

Risk	Likelihood	Impact	Mitigation Strategy
Scope creep due to uncontrolled change requests	High	High	Strict Change Control Board (CCB) process; all CRs reviewed fortnightly
Data quality issues during migration	Medium	High	Data cleansing sprints in Phase 2; automated validation rules in LSMW
User resistance to system adoption	Medium	Medium	Early change management, role-based training, super-user program
Payroll parallel run discrepancies	Low	High	Two-month parallel payroll run before full cutover; reconciliation sign-off
Interface failures (SAP–legacy systems)	Medium	Medium	RFC/BAPI interface testing in Phase 3; fallback manual procedures

Chapter 10 | Proposed Solution & Features

The SAP ERP solution proposed for GreenHarvest Agro Pvt. Ltd. is engineered to replace five disparate legacy systems with a single, integrated platform. The solution introduces the following key capabilities:

10.1 Automation

- Automated MRP execution every Sunday night generates planned orders and purchase requisitions for the upcoming week without manual intervention.
- Automatic Transfer Order creation upon goods receipt eliminates manual warehouse entry and reduces GR-to-bin putaway time from an average of 4 hours to under 30 minutes.
- Payroll automation with the India-specific schema INI ensures zero manual payslip calculation; TDS, PF, and ESI deductions are computed and posted automatically.
- Workflow-based approvals for leave requests (IT2001), purchase requisitions, and production order releases eliminate paper-based approval chains.

10.2 Efficiency Improvements

Process Area	Metric	Pre-SAP (Baseline)	Post-SAP (Target)
Warehouse	Inventory accuracy	88%	99.5%
Payroll	Payroll processing time	4 days	8 hours
Production	Production order cycle time	6.2 days	4.5 days
Procurement	PO creation lead time	3 days	4 hours (auto-PR)
HR	New hire onboarding time	7 days	2 days
Reporting	Monthly MIS report generation	3 days	Real-time (BI)

10.3 Real-Time Tracking

- Batch traceability from raw material lot to finished goods shipment, meeting FSSAI requirements.
- Live production order status visibility (CO03) for plant managers and quality teams.
- Real-time warehouse bin-level inventory (LS26 – Display Warehouse Stock) accessible to all authorised users.
- Employee attendance and leave balances visible to managers via the Manager Self-Service (MSS) portal.

Chapter 11 | Technologies Used

Category	Technology / Tool	Purpose
ERP Platform	SAP ERP ECC 6.0	Core business system for HCM, PP, WM.
Database	SAP HANA (in-memory DB)	High-speed transactional and analytical processing.
Middleware	SAP PI/PO (Process Integration)	Interface between SAP and external systems.
Reporting	SAP BW / SAP Analytics Cloud	MIS dashboards, operational KPIs, management reports.
Self-Service Portal	SAP Fiori / NetWeaver Portal	ESS/MSS for employee and manager self-service.
Data Migration	LSMW / BDC / BAPI	Legacy data migration into SAP master and transactional data.
Testing	HP ALM / JIRA	Test case management, defect tracking during UAT.
Infrastructure	AWS EC2 (SAP-certified)	Cloud hosting for SAP application and DB servers.
Printing	SAP Adobe Document Services	Payslips, warehouse labels, goods receipt notes (GRN).
Mobile WM	SAP Extended Warehouse Mgmt Mobile	Handheld scanner support for warehouse TOs.

Chapter 12 | Unique Selling Points of the Implementation

The GreenHarvest SAP implementation differentiates itself from a generic ERP rollout through several domain-specific design decisions:

USP	Description	Business Value
Farmer-to-Fork Traceability	Custom batch classification in PP links production to field location.	Enables FSSAI compliance and pre-field location certification.
Seasonal Workforce Agility	HCM configured with separate employee subgroups for seasonal hiring.	Reduces payroll costs by 60% during harvest season.
Integrated Quality Hold in WM	QM inspection lots automatically block warehouse storage.	Eliminates quality escape and costly recalls.
Dynamic Bin Sizing in Cold Storage	Storage configured with weight-based bin capacity.	Extends equipment life (10+ yrs) by preventing overloading.
Incentive Payroll from WM Data	Worker productivity (TOs confirmed per day) feeds incentive calculation.	Automates incentive verification; eliminates manual claims.

Chapter 13 | Future Enhancements

The current implementation lays the foundation for a next-generation smart agri-enterprise. The following enhancements are planned on a 3-year technology roadmap:

13.1 AI & Machine Learning Integration

- Demand forecasting using SAP Integrated Business Planning (IBP) with ML-based seasonal demand algorithms, reducing forecast error from 18% to under 6%.
- Predictive maintenance alerts for production equipment (MILL01, PRESS01) using IoT sensor data fed into SAP Asset Intelligence Network (AIN).
- AI-assisted resume screening integrated with SAP SuccessFactors Recruiting for seasonal workforce hiring automation.

13.2 SAP S/4HANA Cloud Migration

- Migration from SAP ECC 6.0 to SAP S/4HANA Cloud (Public Edition) planned in Year 3 of the roadmap.
- Simplified data model (Universal Journal – ACDOCA) will consolidate FI, CO, and MM postings, reducing data redundancy and improving period-close speed by an estimated 70%.
- SAP Fiori 3.0 launchpad will replace the current SAP GUI for all end-user transactions.

13.3 Smart Analytics & Reporting

- SAP Analytics Cloud (SAC) integration will deliver live operational dashboards to C-suite executives on mobile devices.
- Embedded analytics in S/4HANA will replace current BW-based batch reporting with real-time virtual data models (CDS views).
- Sustainability reporting module (SAP ESG) to track carbon footprint per production batch, supporting GreenHarvest's net-zero commitment by 2035.

13.4 Blockchain for Supply Chain Transparency

GreenHarvest plans to pilot SAP Blockchain Services to immutably record each batch's journey from farmer to retail shelf. This will strengthen consumer trust in the 'Certified Organic' label and enable real-time product recall if a quality incident is detected at any point in the supply chain.

Chapter 14 | Conclusion

This project report has presented a detailed and professionally structured SAP ERP implementation blueprint for GreenHarvest Agro Pvt. Ltd., covering the Human Capital Management (HCM), Production Planning (PP), and Warehouse Management (WM) modules. The blueprint encompasses organisational structure design, module-level configuration through SPRO navigation paths, cross-module integration architecture, end-to-end business process flows, and a phased implementation strategy grounded in the SAP ACTIVATE methodology.

The proposed solution directly addresses GreenHarvest's pre-implementation pain points: manual payroll processing, inventory inaccuracies, production scheduling gaps, and the absence of real-time operational visibility. Post-implementation, the organisation is projected to achieve a 99.5% inventory accuracy rate, reduce payroll cycle time from four days to eight hours, cut production order cycle time by 27%, and enable real-time executive dashboards — collectively representing a transformative improvement in operational maturity.

Beyond immediate operational gains, the implementation positions GreenHarvest for a strategic digital future. The roadmap towards SAP S/4HANA Cloud, AI-driven forecasting, IoT-enabled predictive maintenance, and blockchain-based supply chain transparency establishes this blueprint as not merely a technology project, but a comprehensive business transformation initiative.

Learning Outcomes

- Gained deep understanding of SAP ERP organisational structures and their interdependencies across HCM, PP, and WM modules.
- Developed practical knowledge of SAP SPRO configuration paths for enterprise-level system setup.
- Understood cross-module integration mechanisms including payroll-production and warehouse-production data flows.
- Applied the SAP ACTIVATE implementation methodology to design a realistic, phased deployment plan.
- Analysed risk management strategies applicable to large-scale ERP implementations in the agriculture and food-processing industry.

Chapter 15 | References

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End of Report — GreenHarvest Agro Pvt. Ltd. | SAP ERP Implementation Blueprint
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